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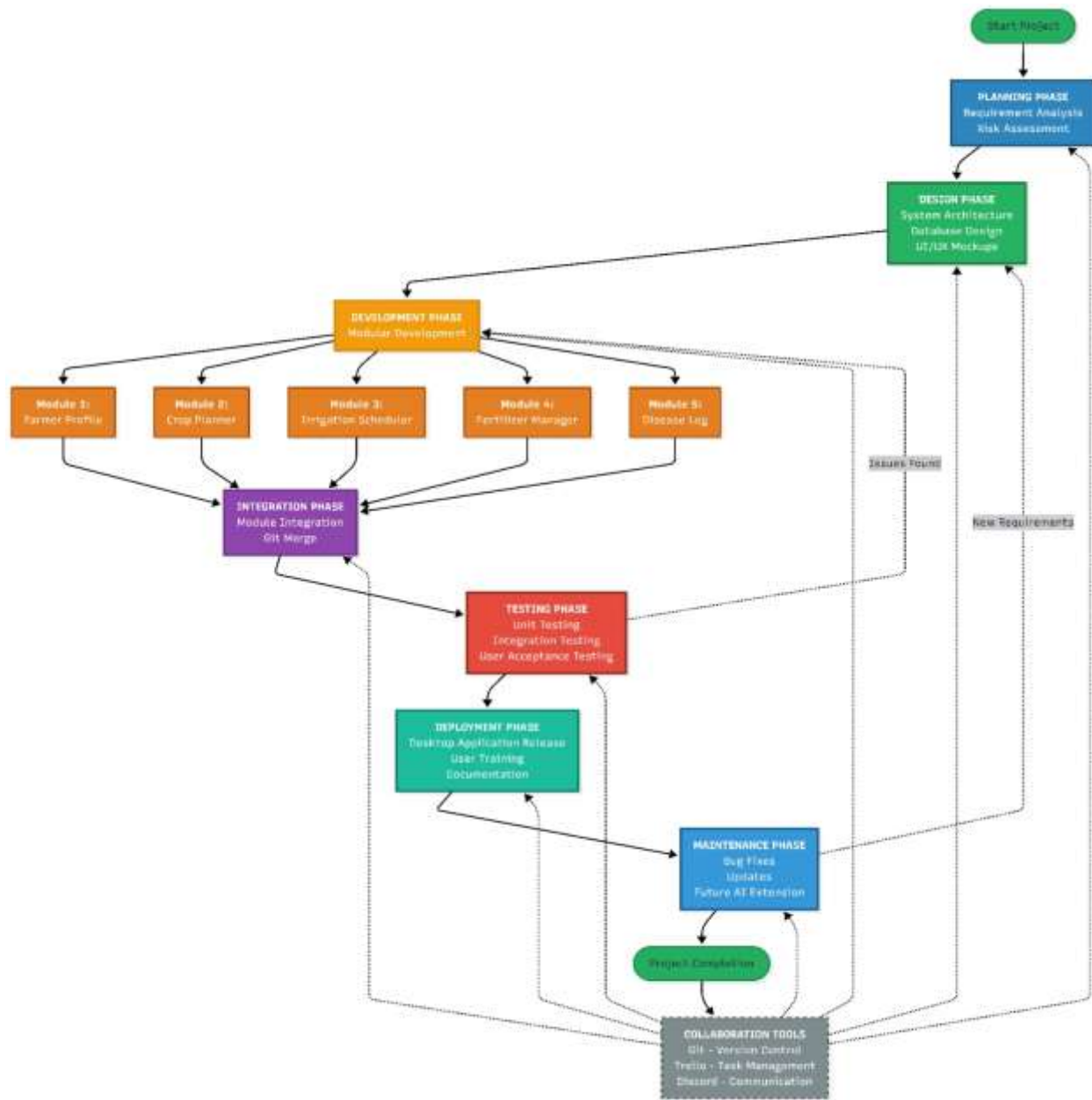
Design of a Use Case Diagram for the Sustainable Farm Decision Support and Crop Advisory System

1. Objective(s)

- To understand the concept of use case diagrams in UML.
- To identify actors and use cases of the farm advisory system.
- To represent system functionality graphically.

2. Problem Analysis

The Sustainable Farm Decision Support and Crop Advisory System allows farmers to input farm details (land size, soil type, previous crops) and receive recommendations for crop selection, irrigation scheduling, fertilizer management, and disease logging. Agriculture officers and researchers can also access the system for monitoring and research purposes.



3. Use Case Diagram

A use case diagram represents the functional requirements of the system. It shows interactions between actors (users or external systems) and the system.

3.1 Key Components

- **Actor:** Farmer, Agriculture Officer, Researcher.
- **Use Case:** Crop recommendation, irrigation scheduling, fertilizer advice, disease logging, etc.
- **System Boundary:** Rectangle enclosing all farm advisory functionalities.
- **Relationships:** Association (solid line), Include (mandatory), Extend (optional).

3.2 Actors and Use Cases

Actor	Description
Farmer	Primary user with basic computer skills
Agriculture Officer	Monitors multiple farms, analyzes performance
Researcher	Uses data for studies and future AI integration

3.3 Use Cases

For Farmer:

- Manage Profile
- Get Crop Recommendation
- Schedule Irrigation
- Get Fertilizer Plan
- Log Disease
- Generate Health Report

For Agriculture Officer:

- View Farm Analytics
- Monitor Multiple Farms

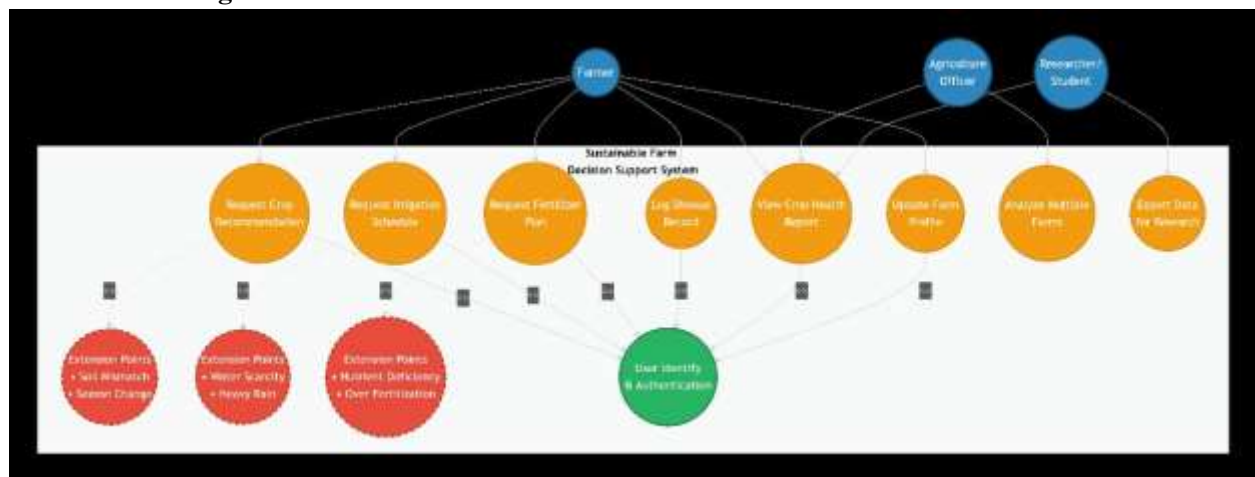
For Researcher:

- Export Data for Research
- View Historical Trends

3.4 Relationships

- **Include:** Authenticate User is included in all actions (if login required).
- **Extend:** Generate PDF Report may extend Generate Health Report.

3.5 Use Case Diagram :



5. User Stories for Farm Advisory System

Format:

As a [user], I want to [action] so that [benefit].

User Stories:

- As a farmer, I want to input my land size, soil type, and previous crops so that the system can give personalized recommendations.
- As a farmer, I want to receive crop suggestions based on season and soil so that I can choose the most profitable crop.
- As a farmer, I want to get irrigation schedules (frequency and quantity) so that I can water my crops efficiently and save water.
- As a farmer, I want to see recommended fertilizer types and amounts so that I can improve crop yield without overusing chemicals.
- As a farmer, I want to log disease symptoms (leaf spots, pest infestations) so that I can track crop health over time.
- As a farmer, I want to generate a health report for my crops so that I can take timely action against diseases.
- As a farmer, I want to update my farm profile easily so that my recommendations stay accurate when conditions change.
- As an agriculture officer, I want to view analytics of multiple farms so that I can identify underperforming areas.
- As a researcher, I want to export farm and disease data so that I can study patterns and improve future AI models.
- As a system, I want to authenticate users before allowing access so that farm data remains private and secure.